
Contents

1	Preliminaries	1
1.1	Homeomorphisms	1
1.2	Algebraic Invariants and the Fundamental Group	3
1.3	Homology	6
2	Calculating Homology and Homotopy	9
2.1	Group Theory: The theoretical minimum	9
2.2	The Theorem of Seifert-van Kampen	10

CHAPTER ONE

Preliminaries

The goal of this text is nothing more but a short **very informal** introduction to algebraic topology, focussed on introducing useful techniques from maths for determining algebraic invariants which are of physical significance.

1.1 Homeomorphisms

In the past, I noticed that words as *homeomorphism*, *diffeomorphism*, *homotopy equivalence* and *homotopy* get mixed up, so I want to make their differences clear and motivate them. I think this becomes especially clear if one considers categories and functors between them to understand those definitions, but we can be more explicit. We start with a topological setup:

Definition 1.1 (Topological Space). A **topology** on a set X is a family $\mathcal{T}$ of subsets of X such that:

$$X \in \mathcal{T} \text{ and } \emptyset \in \mathcal{T}$$

(T2) If I is an arbitrary indexing family and $\forall i \in I : U_i \in \mathcal{T}$, we also have

$$\bigcup_{i \in I} U_i \in \mathcal{T}$$

(T3) For all $U, V \in \mathcal{T}$, $U \cap V \in \mathcal{T}$.

Then, we call $\mathcal{T}$ **topology** on X and $(X, \mathcal{T})$ **topological space**. The sets $U \in \mathcal{T}$ are called **open sets** of X .

Example 1.2. There are many important examples:

- Every metric space is a topological space by taking the topology to be induced by the metric. This makes $\mathbb{R}^n$ and $\mathbb{C}^n$ topological spaces in particular. Since subsets of metric spaces are metric, those are also topological spaces.
- Every set can be turned into a topological space by declaring all possible subsets of that set to be open. This is called the **discrete topology**.
- Similarly, we can take any set X and define a topology by saying X and $\emptyset$ are the only open sets. This is the **indiscrete topology**.

In physics, one often only deals with topologies induced by a metric, so feel

One should think of topological spaces as generalizations of metric spaces. In metric spaces like $\mathbb{R}^n$, open subsets are determined by the Euclidean metric. If one thinks about the properties such open subsets satisfy, (T1)-(T3) are almost immediate. For example, take $\mathbb{R}$ with the usual topology

$$\mathcal{T} := \{U \subseteq \mathbb{R} \mid \forall x \in U \exists \epsilon > 0 : (x - \epsilon, x + \epsilon) \in U\}.$$

For all $x \in \mathbb{R}$, we can take $\epsilon = 1$ and have $(x - 1, x + 1) \subseteq \mathbb{R}$, so $\mathbb{R} \in \mathcal{T}$. For the empty set, the statement is vacuously true, so $\emptyset \in \mathcal{T}$. If $\{U_i\}_{i \in I}$ is a (possibly uncountably infinite) family of open subsets of $\mathbb{R}$ and $x \in \bigcup_{i \in I} U_i$, by definition we have $x \in U_\alpha$ for some $\alpha \in I$. This means that there is $\epsilon > 0$ such that $(x - \epsilon, x + \epsilon) \subseteq U_\alpha$ since all U_i are open. But then we have $(x - \epsilon, x + \epsilon) \subseteq \bigcup_{i \in I} U_i$ by definition of the union. Lastly, given $U_1, U_2 \in \mathcal{T}$ and $x \in U_1 \cap U_2$, we can find $\epsilon_1, \epsilon_2 > 0$ such that $(x - \epsilon_i, x + \epsilon_i) \subseteq U_i$. But then we have $(x - \epsilon, x + \epsilon) \subseteq U_1 \cap U_2$ for $\epsilon = \min\{\epsilon_1, \epsilon_2\}$.

Note two things here: Firstly, being a topological space is really not difficult. However, having an actually useful topology is a whole different story. Secondly, topologies are clearly not unique: We can give $\mathbb{R}^n$ the metric topology, but also the discrete or the indiscrete topology. The obvious choice is of course the first one, which is why we call it canonical.

free to take $\mathbb{R}^n$ as playground for imagination. Now we have sets with new structure, but there is no reason why a function $f : X \rightarrow Y$ should preserve this structure, i.e. there is no reason for $f^{-1}(V)$ to be open in X even if V is open in Y .

Definition 1.3 (Continuity). A function $f : X \rightarrow Y$ between topological spaces is continuous if for all open subsets V of Y , $f^{-1}(V)$ is open in X .

For the usual metric spaces, this agrees with the ϵ - δ -definition of continuity. For two sets A, B , we said that $A \cong B$, or A isomorphic to B , if there is a bijective map between A and B . We want to extend this to topological spaces with their additional structure:

Definition 1.4 (Homeomorphism). A function $f : (X, \mathcal{T}) \rightarrow (Y, \mathcal{T}')$ is called **homeomorphism** if f is bijective and f as well as f^{-1} are continuous. In this case, we write $(X, \mathcal{T}) \cong (Y, \mathcal{T}')$ and say that X and Y are **homeomorphic** or **isomorphic as top. spaces**.

This means when we think about topological spaces, continuous functions with continuous inverses are all we could ask for: They preserve all topological structure. Intuitively, one has to take care of two things: The map cannot tear or crease as it is continuous, and it has to be bijective, so no information should be lost.

Example 1.5. • Every open interval $(a, b) \subseteq \mathbb{R}$ is homeomorphic to $\mathbb{R}$ itself. We can always blow it up or shrink it down as necessary.

- Charts of manifolds are homeomorphisms, so every manifold is **locally** homeomorphic to $\mathbb{R}^n$.
- A point $x \in \mathbb{R}$ can never be homeomorphic to some open interval in $\mathbb{R}$ since there cannot exist a bijection between a finite and an uncountably infinite set. Similarly, $\mathbb{S}^1$ cannot be homeomorphic to an open interval as $\mathbb{S}^1$ is compact while open intervals are not.
- Size is not a topological property: All open balls of $\mathbb{R}^n$ are homeomorphic by translation $x \mapsto x + y$ or dilation $x \mapsto \lambda x$.

We conclude this digression with probably the most intricate definition:

Definition 1.6 (Topological Embedding). Let S, X be topological spaces and $f : S \rightarrow X$ be a function. We call f **topological embedding** if f is injective and the restriction

$$\tilde{f} : S \rightarrow f(S) \subseteq X$$

is a homeomorphism.

Example 1.7. • The graph of the parabola given by

$$F : \mathbb{R} \rightarrow \mathbb{R}^2, x \mapsto (x, x^2)$$

is an embedding: It is injective everywhere and on the image of F , we can

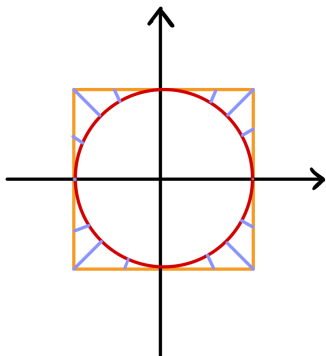


Figure 1.1: A circle can be blown up to a square and a square can be shrunk down to a circle. This is a homeomorphism.

One should think of an embedding as having a true copy of S inside of X such that the topology of S as a subset of X (which is obviously induced by the topology on X) agrees with the original topology of S as a topological space.

1.2. ALGEBRAIC INVARIANTS AND THE FUNDAMENTAL GROUP

use the projection $(x, y) \mapsto x$ as inverse map. Both F and the projection are clearly continuous.

- Every topological manifold of dimension n can be embedded into some $\mathbb{R}^N$ for N large enough. Hence every manifold is a submanifold (*Whitney embedding theorem*).
- The lemniscate (figure 8), given as

$$G : [0, 1] \rightarrow \mathbb{C}, t \mapsto \exp(2\pi i t),$$

is not an embedding as it fails to be injective: $G(0) = G(1)$. Removing the problematic self-intersection,

$$G' : [0, 1) \rightarrow \mathbb{C}$$

is still not an embedding as the metric topology on $\mathbb{C}$ does not agree with the topology of $G'([0, 1))$. This is a good example for why embeddings are sometimes a bit tricky. The good news is that restriction on any smaller interval $[0, a)$ with $a < 1$ will also solve this problem, turning the open lemniscate into a topological subspace of $\mathbb{C}$.

We can now collect the first result about the problem at hand.

Proposition 1.8. The map

$$F : \mathbb{S}^1 \times \mathbb{S}^1 \rightarrow \mathbb{R}^3 \quad (1.1)$$

$$(u, v, w, z) \mapsto (u + w, v, z) \quad (1.2)$$

fails to be an embedding.

Proof. Consider the complex numbers $(1, 0)$ and $(-1, 0)$ which are clearly on the unit circle $\mathbb{S}^1 \subseteq \mathbb{C}$. Then, we have:

$$F((1, 0), (-1, 0)) = (0, 0, 0) = F((-1, 0), (1, 0)),$$

and hence F is not injective. $\square$

1.2 Algebraic Invariants and the Fundamental Group

Homeomorphisms are great. They preserve all possible topological structure. However, they are also often very difficult or do not exist at all between two given spaces. Therefore, we want to look for means to compare topological spaces that are less strict, but still fine enough to be actually useful. In practice, we are looking for algebraic invariants: Informally speaking, algebraic invariants are algebraic objects we can assign to a topological space which are preserved under some kind of map. This makes it possible to characterize spaces in regard to some designated property, stripping the space of all unnecessary data for the problem at hand. Those invariants often come in the form of special numbers (e.g. Euler characteristic) or groups.

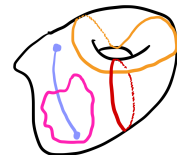


Figure 1.2: Some open and closed paths in a space homeomorphic to a torus.

Definition 1.9 (Path). Let X be a topological space. A **path** in X is a continuous map

$$\gamma : [0, 1] \rightarrow X.$$

We say that γ is closed if $\gamma(0) = \gamma(1)$, or equivalently, γ is a continuous map $\mathbb{S}^1 \rightarrow X$.

Our goal is to detect fundamental properties of spaces with paths. One can immediately think of holes: Somehow, the main difference between the torus and a sphere should be that the torus has a hole, while the sphere does not. Which kind of map does preserve at least holes?

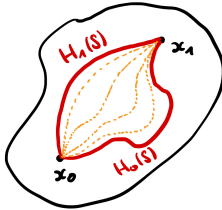


Figure 1.3: A continuous path homotopy with fixed endpoints.

Definition 1.10 (Path homotopy). Two paths $\gamma, \omega : [0, 1] \rightarrow X$ are **path-homotopic** if $\gamma(0) = \omega(0) =: x_0$, $\gamma(1) = \omega(1) =: x_1$, and there is a continuous map

$$H : [0, 1] \times [0, 1] \rightarrow X$$

with:

- $H(s, 0) = H_0(s) = \gamma(s)$
- $H(s, 1) = H_1(s) = \omega(s)$
- $H(0, t) = H_t(0) = x_0$
- $H(1, t) = H_t(1) = x_1$

We call H a **path homotopy** and write $\gamma \simeq \omega$.

Example 1.11. Consider the punctured plane $\mathbb{R}^2 \setminus \{0\}$. The paths $\gamma_1(t) := (\cos \pi t, \sin \pi t)$ and $\gamma(t) := (\cos \pi t, n \cdot \sin \pi t)$ are path-homotopic for all $n > 0$. One just connects γ_1 and γ_2 by straight lines and slides all points along them. This is a homotopy. However, γ_1 is never homotopic to $\gamma_3(t) := (\cos \pi t, -\sin \pi t)$, as every homotopy would have to pass through the origin, which was removed by assumption.

This should give a hunch about why homotopies discern certain paths. We are ready to define the first algebraic invariant:

Definition 1.12 (Fundamental Group). Let X be a topological space. The **fundamental group** of X with base point $x_0 \in X$, written $\pi_1(X, x_0)$, is given by homotopy classes of closed paths $\mathbb{S}^1 \rightarrow X$.

So if two paths have a homotopy between them, the fundamental group does treat them as the same path. Why is this a group? The proof is a bit annoying, but we do some heuristics: Two paths can be concatenated to a new path if the endpoint of the first path is the starting point of the second path. The inverse of a path is given by going backwards, i.e. reparametrizing $t \mapsto -t$, and the identity is just standing around, i.e. the constant path $\gamma(t) = x_0$. We still need to know when two spaces are "identical enough" to have the same fundamental group. Previously, we defined a homotopy for paths. Now we consider general homotopies between continuous maps,

1.2. ALGEBRAIC INVARIANTS AND THE FUNDAMENTAL GROUP

which are almost identical apart from the fact that we do not demand fixed endpoints.

Definition 1.13 (Homotopy Equivalence). Two topological spaces X and Y are said to be **homotopy equivalent** if there are maps $f : X \rightarrow Y$ and $g : Y \rightarrow X$ such that $f \circ g \simeq \text{id}_Y$ and $g \circ f \simeq \text{id}_X$, where $\simeq$ means that there exists a homotopy (not necessarily a path-homotopy) between the left and the right side. We write $X \simeq Y$.

So this is our criterion for the fundamental group to be equal:

Proposition 1.14. Let X, Y be topological spaces, $x_0 \in X, y_0 \in Y$. The fundamental groups are isomorphic

$$\pi_1(X, x_0) \cong \pi_1(Y, y_0)$$

if one of the following conditions holds:

- X and Y are homeomorphic.
- X and Y are homotopy equivalent.

Remark 1.15. In general, it is very difficult to calculate the fundamental group of a space from first principles. One famous example is the very annoying proof that $\pi_1(\mathbb{S}^1) \cong \mathbb{Z}$. As physicists, it is somewhat nice that we can use our intuition to infer some fundamental groups without giving a full proof.

Example 1.16. Whenever it is not of concern, we will suppress the basepoint in the notation of π_1

- The real space $\mathbb{R}^n$ has trivial fundamental group for all n . We call such spaces **simply connected**. Note: *Even though $\pi_1(\mathbb{R}^n) \cong \pi_1(\mathbb{R}^m)$ for $n \neq m$, there can never be a homeomorphism $\mathbb{R}^n \rightarrow \mathbb{R}^m$ for $n \neq m$. This is an example where the fundamental group is not enough to predict topological equivalence.*
- $\pi_1 \mathbb{S}^1 \cong \mathbb{Z}$: This is seen intuitively by realizing that there are only two fundamentally different closed paths in $\mathbb{S}^1$: One can take the constant path, which is isomorphic to $0 \in \mathbb{Z}$. For all other paths, the only sensible difference is how often and in which direction one traverses the circle. Direction is mapped to the sign of the integer, and the amount of closed loops to the modulus.
- $\pi_1(\mathbb{S}^n) \cong \{0\}$ for all $n > 1$. This is something to remember well: All spheres in dimension 2 and higher are simply connected. For $\mathbb{S}^2$, one can imagine a rubber band sliding on a basketball to see the homotopy connecting every path to the trivial one.
- $\pi_1(\mathbb{S}^1 \times \cdots \times \mathbb{S}^1) \cong \mathbb{Z} \times \cdots \times \mathbb{Z}$: Remember that $\mathbb{S}^1 \times \mathbb{S}^1$ is the torus, so there are two fundamental paths in the torus and hence the fundamental group consists of two copies of the integers, one for each path.

Every homeomorphism gives rise to a homotopy equivalence as the first one is much stronger, like we discussed. But when are homotopy equivalences not homeomorphisms? This occurs often because of failure of bijectivity. One of the funniest examples is that the letter X and the letter Y are homotopy equivalent, but not homeomorphic. The equivalence is given by shortening one of the spikes of X to the center of the letter X. This maps the letter X to Y. We map Y to X by mapping the three spikes of Y to three out of four spikes of X. Performing these moves in succession is just the identity, we end up with either X or Y. But the maps are very clearly non-surjective and non-injective.

Think of $\mathbb{Z}$ as abelian group under addition: The neutral element is 0.

- One shall not imagine all fundamental groups to be nicely abelian. Take the connected sum of two circles, $\mathbb{S}^1 \vee \mathbb{S}^1$. The fundamental group is $\mathbb{Z} \star \mathbb{Z}$, where $\star$ denotes the free product of groups. We will introduce this later. For now, note that the free product is very far from being abelian, i.e. commutative.

We will need another definition already mentioned: The paths I called fundamental (e.g. traversing $\mathbb{S}^1$ once, or traversing meridian and longitude of the torus once) are called **generators** of the fundamental group, since all other paths can be obtained by walking that path multiple times and/or reversing direction. Thinking abstractly, the generators of $\pi_1(\mathbb{S}^1) \cong \mathbb{Z}$ are just $\pm 1 \in \mathbb{Z}$.

There is indeed a connection between the fact that $\mathbb{S}^1$ has two generators differing only in sign and also two orientations induced by tangent vector fields.

1.3 Homology

If π_1 is too complicated, we could try to simplify it as a group. π_1 is pretty clearly non-abelian, so $\gamma_1 * \gamma_2 \neq \gamma_2 * \gamma_1$ in general. What happens if we just force all non-commuting paths to commute? Unfortunately, this is a bit more involved than defining π_1 , but in return we get directly all homology groups, not only the first one. We denote the n -simplex by Δ^n . This is just a hypertriangle, if one wants to imagine it like that: For $n = 1$, we have a point. For $n = 2$, we have a straight line, i.e. the interval $[0, 1]$. For $n = 3$, we have a filled-out triangle. For $n = 4$, we have a filled-out tetrahedron, and so on. Given a top. space X , a **singular n -simplex** is a continuous map

$$\sigma : \Delta^n \rightarrow X.$$

This is done in group theory by abelianizing: We take the quotient of the non-abelian group by the subgroup generated by all elements of the form $aba^{-1}b^{-1}$.

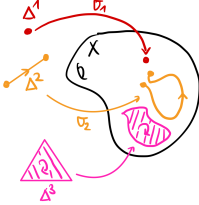


Figure 1.4: Different singular simplices are mapped into a top. space X .

This means we just wiggle our n -simplex in our top. space without tearing or creasing. There is a map $\partial_n : \Delta^n \rightarrow \Delta^{n-1}$ for any n which deletes the interior of the n -simplex, leaving a sum of $n - 1$ -simplices. We call this the **face operator**.

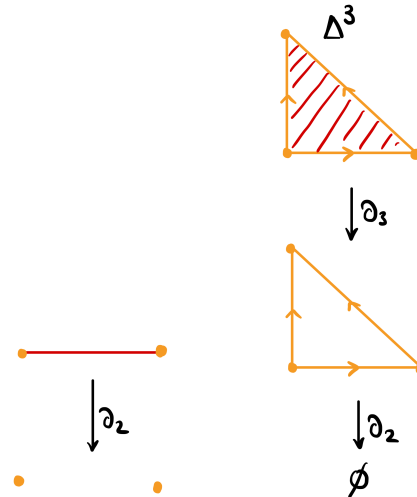


Figure 1.5: The face operator deletes the interior and leaves only a boundary. You can observe that a boundary has no boundary itself, so $\partial_n \circ \partial_{n-1} = 0$ for all n .

1.3. HOMOLOGY

Now we want to be able to consider many n -simplices at once, and we also want that two simplices which differ only in orientation cancel each other out, just like $-1 + 1 = 0$. Thus, we just allow formal sums of singular simplices with coefficients in $\mathbb{Z}$. This yields a space which we call the n -th chain C_n . The map ∂_n acts linearly on each chain, yielding a so-called boundary operator $d_n : C_n \rightarrow C_{n-1}$ for each n .

Definition 1.17 (Homology). The n -th homology group is given by

$$H_n(X) := \ker(\partial_n) / \text{im}(\partial_{n+1}).$$

What does this mean? We take the kernel of the n -th boundary map, so everything that gets mapped to zero by ∂_n . Thinking in $n = 2$, so paths, this means that paths without any boundary are in the kernel. This are precisely the closed paths, since an open path has its two endpoints as boundary. What is the image of ∂_{n+1} ? This consists of paths which are boundaries of "deformed" triangles in X . Take for example the torus: The path wrapping around the meridian cannot be a boundary since there is no room in between the hole to have a 3-simplex with the path as boundary. So this element is in $H_2(T)$. Same goes for the longitudinal path. The image consists of paths not wrapping around one of the holes, these are considered trivial and removed from $H_2(T)$. So we have two generators in $H_2(T)$ of infinite order. This yields heuristically

$$H_2(T) \cong \mathbb{Z} \times \mathbb{Z}.$$

In higher dimensions, there is not really a nice intuition anymore. However, notice the difference to π_1 : We do not demand any base point of our space. In fact, if one is concerned with counting holes of spaces, $H_n(X)$ is the way to go. Let us remark a quick sanity check for our goal:

Theorem 1.18 (Hurewicz Isomorphism Theorem). For any top. space X , the map

$$\pi_1^{\text{ab}}(X) \rightarrow H_1(X)$$

is a group isomorphism where π_1^{ab} denotes the abelianized fundamental group.

This tells us many important things: Firstly, the first homology group contains less information than the fundamental group since many things can be lost by abelianizing. This makes homology easier, but also rougher. Secondly, if we know π_1 , we get many H_1 for free: If they are abelian, they stay just the same. Lastly, the Hurewicz map also exists for higher homotopy groups π_n , but it fails to be an isomorphism for all but $n = 1$. Not everything is that nice...

A little reminder from MfP2: The kernel of a map consists of everything the map sends to 0. Take for example the projection $p : \mathbb{R}^2 \rightarrow \mathbb{R}^1$ by collapsing the y -axis. The kernel of that map is the whole y -axis, so $\ker(p) = \mathbb{R}$. Also another reminder: Taking the quotient A/B of a group A by a subgroup B (read $A \bmod B$) means to form a new group with elements from A , but where all elements of B are set to 0. An example would be $\mathbb{R}/\mathbb{Z} \cong \mathbb{S}^1$. Think of gluing the real line together at all integers, so one gets infinitely many copies of $\mathbb{S}^1$, which is, of course, homeomorphic to $\mathbb{S}^1$.

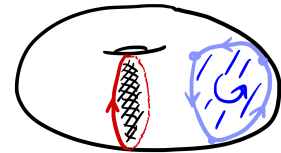


Figure 1.6: The torus has a singular 2-simplex visible in red which cannot possibly bound a singular 3-simplex because of the hole. There is also another such simplex along the longitude. The blue singular 2-simplex on the other hand is a boundary and does not contribute to homology.

There are some funny groups that vanish completely if you try to abelianize them. They are called **perfect groups**, although a lecturer of mine suggested the better fitting term Kain-ian group. One example is the alternating group A_5 , a subgroup of the symmetric group on five letters containing only odd permutations.

CHAPTER TWO

Calculating Homology and Homotopy

2.1 Group Theory: The theoretical minimum

Unfortunately, we need some quite abstract concepts from group theory to understand how to calculate fundamental groups effectively. I will try to keep it at an absolute minimum, just enough to understand the fundamental theorem of this section. Our goal is to define the amalgamation of two groups. We start with a construction: Begin with a set S of n letters $\{s_1, \dots, s_n\}$ and add their formal inverse to this set, so we get $\{s_1, s_1^{-1}, \dots, s_n, s_n^{-1}\}$. Now add all possible words you get by arranging those $2n$ letters freely. You are not allowed to permute, so $s_1 s_2 \neq s_2 s_1$. You are allowed to add a letter multiple times, like $s_1 s_1 s_1 s_1 s_1 = s_1^5$ is a word. The set obtained in this way is denoted $\mathcal{F}_n$. Define a group operation on those words by just writing them left to right as one word, e.g. on $\mathcal{F}_3$ we could write

$$aba \circ cac = abacac.$$

Define the group inverse of s_i as the formal inverse s_i^{-1} such that one can cancel adjacent inverse letters in words, e.g. $abaa^{-1} = ab$. But note that aba^{-1} , for example, cannot be simplified. The neutral element is the empty word. Now reduce all words as far as possible, i.e. cancel all adjacent inverses. This defines a group structure on $\mathcal{F}_n$ and we call $\mathcal{F}_n$ the **free group on n generators**.

Remark 2.1. As far as groups can be nice, free groups are the exact opposite of a nice group: Even the free group on one generator is infinitely big as it contains all exponents of a . Also, for $n > 2$ it is never abelian.

Interesting about free groups is that they can be used to smash two groups together into a new group without caring about how the two groups interact with each other.

Definition 2.2 (Free Product). Let G, H be two groups. The free product $G \star H$ is defined to be the free group generated by all elements of G and H after performing the following reductions:

Remove any instance of the identity element of G or H .

- Replace any pairs of the form $g_i g_j$ by their product in G and any elements of the form $h_i h_j$ by their product in H .

This means that we just form words with elements in G and H , treating those simply as letters. We allow absolutely no interaction between G and H in the free product, we can only reduce inside of G and inside of H , e.g. $hh'ghg' = h'ghg$ where $h \circ h' = h''$ in H .

We have one big problem: Given two or even more groups and forming their free product, how do we keep track what can actually be reduced in each of the factors and what not? We need a nice, concise notation to keep track of words and relations. Take a set of letters $\{a_1, \dots, a_n\} =: S$ and denote by $\tilde{S}$ the set of all letters and their formal inverses. Let R be a set consisting of certain words of the letters in $\tilde{S}$. Define

$$\langle S \mid R \rangle$$

to be a quotient of the free group $\mathcal{F}_n$ where all words in R are allowed to be reduced.

Definition 2.3 (Presentation). Let G be a group. If there exist some S and R as above such that

$$G \cong \langle S \mid R \rangle,$$

we call $\langle S \mid R \rangle$ a **presentation** of G . The set R is called **relation set**.

Note that presentations are non-unique. It is even worse: Given a finitely generated group G and two words in the letters of G , it is algorithmically undecidable whether they represent the same element of G or not. This is the famous *word problem*.

Example 2.4.

- The easiest one is $\mathcal{F}_n \cong \langle \{a_1, \dots, a_n\} \mid \emptyset \rangle$. There are no relations.
- We can now attack our problem: Let $G \cong \langle S_G \mid R_G \rangle$ and $H \cong \langle S_H \mid R_H \rangle$ be presentations of G and H . Then

$$G * H \cong \langle S_G \cup S_H \mid R_G \cup R_H \rangle$$

is a presentation of the free product of G and H .

- The integers have presentation $\langle a \mid \emptyset \rangle$ and are hence isomorphic to $\mathcal{F}_1$.
- The cyclic groups have presentation $\mathbb{Z}/n\mathbb{Z} \cong \langle a \mid a^n \rangle$.

Almost done! We have everything to define the amalgamated free product.

This seems more abstract than it actually is: We have maps $H \rightarrow G_i$. For simplicity, assume they are injective, so they identify some part of G_i with H . Then we take the free product of G_1 and G_2 but we make sure that the part of G_1 we identified with H agrees with the counterpart in G_2 . That is all there is to this construction.

Definition 2.5 (Free Product with Amalgamation). Given three groups G_1, G_2, H with presentations $\langle S_{G_i} \mid R_{G_i} \rangle, \langle S_H \mid R_H \rangle$ and two group homomorphisms $\iota_i : H \rightarrow G_i$, we define the **free product of G_1 and G_2 amalgamated with H** as the group

$$G_1 \star_H G_2 := \langle S_{G_1} \cup S_{G_2} \mid R_{G_1}, R_{G_2}, \forall h \in H : \iota_1(h) = \iota_2(h) \rangle. \quad (2.1)$$

And with this, we do some interesting stuff.

2.2 The Theorem of Seifert-van Kampen

We want to calculate π_1 under some reasonable conditions on our topological space. The idea we follow here is to split the space into easier spaces which we understand well to obtain π_1 of the full space. This will prove to be especially useful for top. spaces given as fundamental diagrams, i.e. "glueing squares".

Theorem 2.6 (Seifert-van Kampen). Let $(X, \mathcal{T})$ be a path-connected top. space, $X = X_1 \cup X_2$ for some $X_1, X_2 \in \mathcal{T}$ and $x_0 \in X_0 := X_1 \cap X_2$. Furthermore, let X_0, X_1 and X_2 be also path-connected. Then there is an isomorphism of fundamental groups:

$$\pi_1(X_1, x_0) \star_{\pi_1(X_0, x_0)} \pi_1(X_2, x_0) \cong \pi_1(X, x_0). \quad (2.2)$$

The nice thing about this theorem is that we can prove many things with nice diagrams. The not so nice part is that you really need all the assumptions. This theorem is horribly wrong if one of those spaces is not path-connected or open.

Example 2.7. • Consider again the connected sum of two circles (figure 8): We cover the left and the right circle respectively with circles with small overlap, see the figure. We know that $\pi_1(\mathbb{S}^1) \cong \mathbb{Z}$, and the overlap retracts onto a point, so $\pi_1(X_0, x_0) \cong \{0\}$. This is a proof of our previous statement

$$\pi_1(\mathbb{S}^1 \vee \mathbb{S}^1) \cong \mathbb{Z} \star_0 \mathbb{Z} \cong \mathbb{Z} \star \mathbb{Z}.$$

- The previous result is very useful: Looking at the picture, you can see that the punctured Klein flask $K \setminus \{x\}$ is a retract of $\mathbb{S}^1 \vee \mathbb{S}^1$ and hence $\pi_1(K \setminus \{x\}) \cong \mathbb{Z} \star \mathbb{Z}$.

We want to illustrate the general principle of how to use Seifert-van Kampen with fundamental polygons with an example.

Proposition 2.8. The fundamental group of the Klein bottle K is given by

$$\pi_1(K, x_0) \cong \mathbb{Z} \rtimes \mathbb{Z} \cong \langle a, b \mid abab^{-1} \rangle.$$

Proof. We follow the picture in the margin:

1. Start by drawing the fundamental polygon of the Klein bottle and assign to each path generating the fundamental group a name, here γ_1 and γ_2 .
2. Choose X_1 to be a small open disk in the middle of the polygon. Choose X_2 to be everything but a smaller closed disk in the middle, so that X_2 is open too. Choose some x_0 in the intersection of X_1 and X_2 .
3. We embed the subsets $\iota_1 : X_1 \hookrightarrow X$ and $\iota_2 : X_2 \hookrightarrow X$. This induces homomorphisms $\iota_{1,*}, \iota_{2,*} : \pi_1(X_i) \rightarrow \pi_1(X)$. These homomorphisms are the ones we need for the amalgamated product. Now choose a path in X_0 going through x_0 and traverse the fundamental polygon once in the chosen direction. When you first pass a generator, write the generator down with a positive sign if you go in the same direction or with a negative sign else.

This is all we need. Now we calculate the needed fundamental groups: The open disk is simply connected, so

$$\pi_1(X_1) \cong \{0\}$$

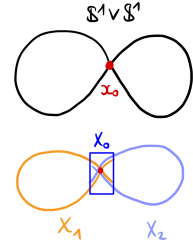
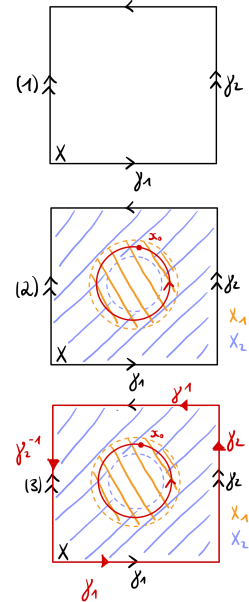


Figure 2.1: We take X_1 and X_2 to be circles with little overlapping open tails. Those open tails retract onto the circles and do therefore not influence the fundamental group. The overlap is a small, open, x-shaped set which retracts onto a point and hence has trivial fundamental group.



Note how the notation for amalgamation $G_1 \star_H G_2$ suppresses very important information as the amalgamation depends heavily on the maps $\iota_i : G_i \rightarrow H$ not mentioned in this notation.

Actually, if this were not the case, the fundamental group of the Klein bottle would agree with that of the torus.

and we get no relations from this one. . The overarching set X_2 has two generators, so it is the free group on two generators

$$\pi_1(X_2) \cong \langle \gamma_1, \gamma_2 \mid \emptyset \rangle.$$

Now the interesting part: X_0 is a so-called open annulus which retracts onto $\mathbb{S}^1$ and hence

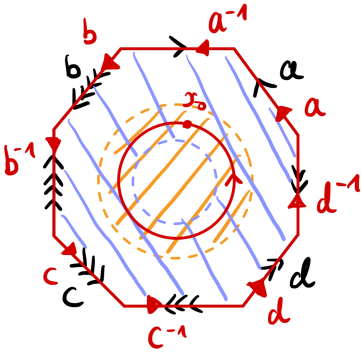
$$\pi_1(X_0) \cong \mathbb{Z}$$

. But we have to be careful: The amalgamation forces the embedded red path to agree with the two generators on the boundary. Traversing the boundary once, this forces $\gamma_1 \gamma_2 \gamma_1 \gamma_2^{-1} = 1$, and hence

$$\pi_1(K) \cong \{0\} \star_{\mathbb{Z}} \mathcal{F}_2 \cong \langle \gamma_1, \gamma_2 \mid \gamma_1 \gamma_2 \gamma_1 \gamma_2^{-1} \rangle.$$

□

Now we want to tackle our real problem.



Proposition 2.9. The fundamental group of the top. space given by the fundamental hexagon on the left is

$$\pi_1(X) \cong \mathbb{Z} * \mathbb{Z} * \mathbb{Z} * \mathbb{Z} \cong \mathcal{F}_4. \quad (2.3)$$

Proof. Same game as above, choose X_1 , X_2 and X_0 accordingly. In this case, $\pi_1(X_2)$ is the free group on four generators, $\langle a, b, c, d \mid \emptyset \rangle$. The fundamental group of X_1 and X_0 are still the same. The embedding of a path in X_0 traverses generators in the following way:

$$aa^{-1}bb^{-1}cc^{-1}dd^{-1}.$$

By Seifert-van Kampen, we obtain:

$$\pi_1(X) \cong \{0\} \star_{\mathbb{Z}} \mathcal{F}_4 \cong \langle a, b, c, d \mid aa^{-1}bb^{-1}cc^{-1}dd^{-1} \rangle. \quad (2.4)$$

But we are always allowed to reduce mutually inverse elements, so the relation reduces to $1 = 1$, the trivial relation. So we just get the free group on four generators again. □

Note that this means that our space is homotopy equivalent to the wedge sum of four circles $\mathbb{S}^1 \vee \mathbb{S}^1 \vee \mathbb{S}^1 \vee \mathbb{S}^1$.